

Department of Computer Science



Federal Urdu University
of Arts, Science and Technology

ARTIFICIAL INTELLIGENCE

ASSIGNMENT

Topic: Smart Vacuum Cleaner Agent

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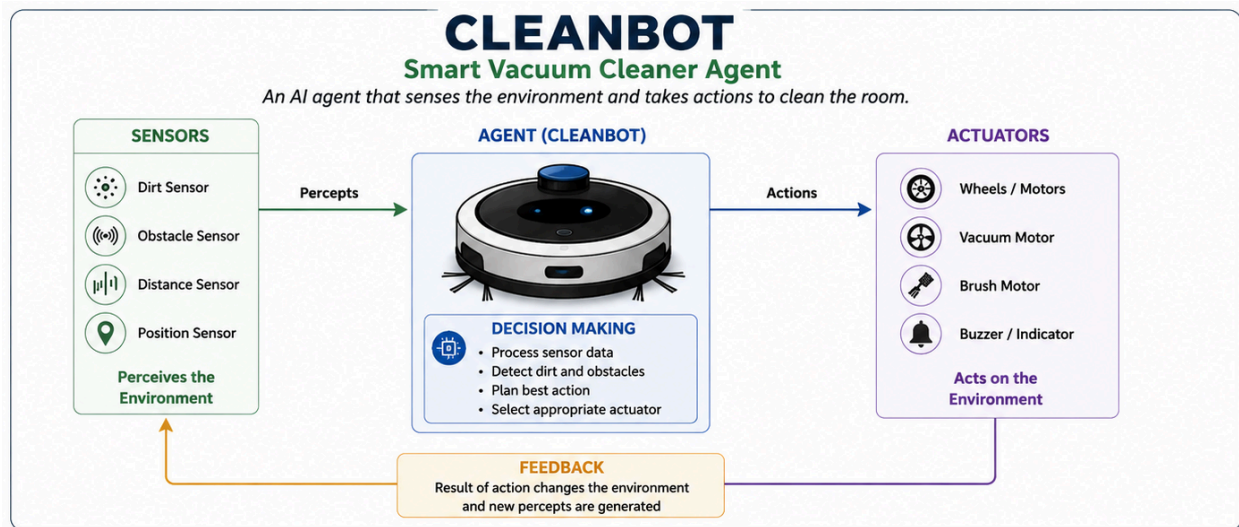
Submitted To: Ms. Uzma Afzal

CLEANBOT

CleanBot is an intelligent vacuum cleaner robot, which can move around the room, avoid obstacles and vacuum dirt on the floor, and make decisions to achieve the goal of keeping an area clean.

Graphical Representation

The basic processing workflow of the CleanBot can be demonstrated as a simple cycle, where the process begins with sensors acquiring information about the environment, after which the agent interprets the information, makes decisions, and the actuators carry out the decisions to ensure an efficient cleaning process and avoid collisions.



Agent Formulation

The CleanBot agent is an entity that uses its various sensors to receive information about the environment. The agent then performs certain functions after analyzing the information it receives.

Environment

This refers to a room or house in which the CleanBot works. It comprises the floor, dirt, furniture, walls, and other obstacles.

Sensors

The CleanBot uses several sensors:

Dirt Sensor: This helps the robot detect dirt on the floor.

Obstacle Sensor: This helps the robot detect obstacles such as furniture and walls.

Distance Sensor: This helps the robot monitor distance.

Position Sensor: This helps the robot determine its position.

Actuators

The robot employs various modules to perform certain tasks:

Wheels: These enable the robot to move in different directions.

Vacuum Motor: This sucks dust and dirt from the floor.

Brush Motor: This helps clean the floor.

Buzzer/Indicator: This helps indicate some status messages or alerts.

Types of Environment

Property	Type
Observability	Partially Observable
Determinism	Stochastic
Episodes	Sequential
Change	Dynamic
Values	Continuous
Agents	Multi-Agent

PEAS Framework

Component	Description
P – Performance	Clean floor, efficient cleaning
E – Environment	Rooms, floors, furniture
A – Actuators	Wheels, brushes, vacuum motor
S – Sensors	Dirt, obstacle, and distance sensors